

POWERFACTORY

PowerFactory 2024

Technical Reference

Q(P)-Curve

IntQpcurve

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

Publisher:
DIgSILENT GmbH
Heinrich-Hertz-Straße 9
72810 Gomaringen / Germany
Tel.: +49 (0) 7072-9168-0
Fax: +49 (0) 7072-9168-88
info@digsilent.de

Please visit our homepage at:
https://www.digsilent.de

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1 General Description

The Q(P)-Curve object (*IntQpcurve*) is used to define for synchronous machines, static generators, station controllers and for loads the behaviour of the reactive power depending on the active power.

The curve can be entered in generator orientation or load orientation.

2 Configuration

The Q(P)-Curve object can be configured at the Configuration page as follow:

- Values given in Mvar/MW
- Values given in p.u.

When the characteristic is defined in p.u. the corresponding rated apparent power is used to convert the characteristic values from p.u. to Mvar and MW.

$$Q[i] = P[i] \cdot S_r \cdot ngnum \tag{1}$$

(2)

where S_r is the rated power of the machine, load or station controller. The $ngnum$ is the no. of parallel units e.g. for static generators or no. of parallel machines for synchronous machines.

Please note, that the curve is defined for one machine. So when the no. of parallel units/machines is greater than one, the characteristic is scaled by the no. of parallel units/machines ($ngnum$) also in case of the Q(P)-Curve is defined in Mvar/MW.

3 Orientation

The characteristic can be specified:

- Generator Oriented
- Load Oriented

3.1 Generator Oriented

When the curve is defined as generator oriented, the entered P, Q values are fed-in active and reactive power values. In case of a generator orient Q(P)-Curve is used e.g. for loads, see corresponding section 4.1 or 4.2.

3.2 Load Oriented

When the curve is defined as load oriented, the entered P, Q values are active and reactive power consumption values. In case of a load orient Q(P)-Curve is used e.g. for a generators, see corresponding section 4.1 or 4.2.

4 Curve definition

The Q(P)-Curve can either be entered as look-up table or polynomial function.

4.1 Look-up Table

The Q(P)-Curve can be defined as look-up table by entering Q, P pairs.

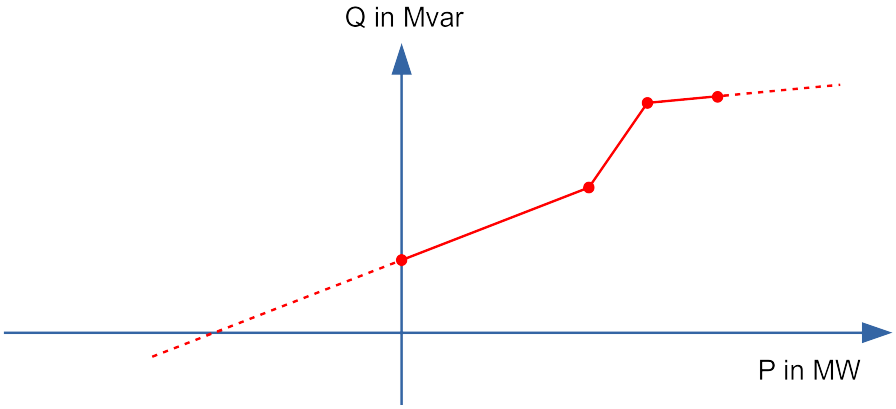


Figure 4.1: Example of a look-up table

When the active power is greater than the last point or lower than the first point, the curve is linear interpolated (see dashed lines).

In case of the orientation is different, e.g. the curve is defined as load oriented and used for generators, the P,Q pairs are correspondingly negated (P and Q). The following figure shows an example, the red points are the entered P, Q values.

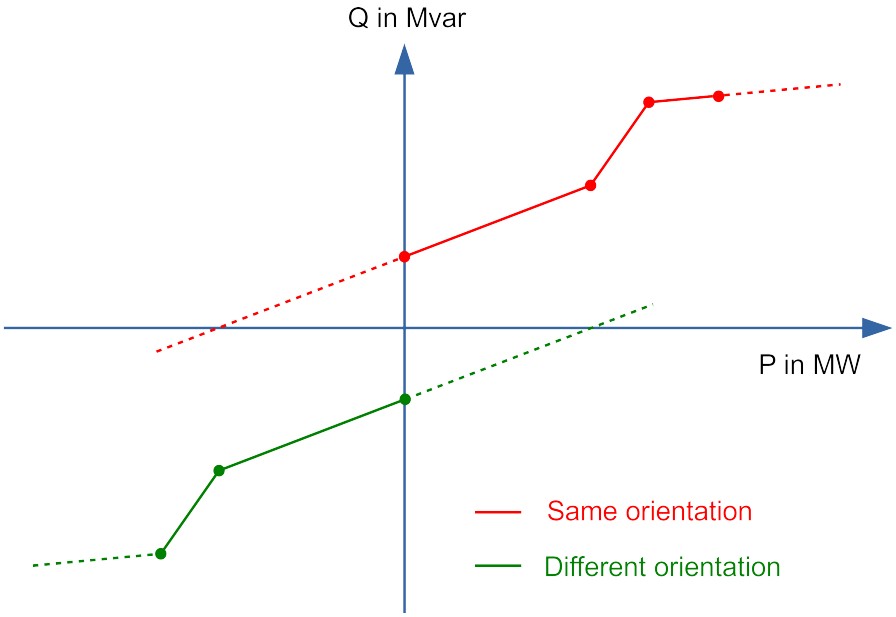


Figure 4.2: Example of a look-up table with different orientation

4.2 Polynomial

The Q(P)-Curve can be defined as polynomial function

$$Q = aP \cdot P^2 + bP \cdot P + cP \tag{3}$$

where

- *aP*: Coefficient aP
- *bP*: Coefficient bP
- *cP*: Coefficient cP

and P is the active power in MW of one machine.

In case the Q(P)-Curve is defined in p.u. values the following equation is used:

$$Q_{pu} = aP \cdot P_{pu}^2 + bP \cdot P_{pu} + cP \tag{4}$$
$$Q = Q_{pu} \cdot S_r \tag{5}$$

The calculated reactive power Q_{pu} is than multiplied by the rated power of the machine (S_r) to get the reactive power for one machine.

If the no. of parallel units/machines is greater than one, the reactive power Q is multiplied by $ngnum$ to get the total reactive power.

Note: When the Q(P)-Curve is defined as load oriented and used e.g. for a synchronous machine or static generator the definition of the curve is considering the different orientation as follow:

$$Q = d_o \cdot aP \cdot P^2 + bP \cdot P + d_o \cdot cP \tag{6}$$

d_o is -1 in case of the orientation is different e.g. curve is load oriented but used for generators or curve is generator oriented but used for loads. See following figure for an example:

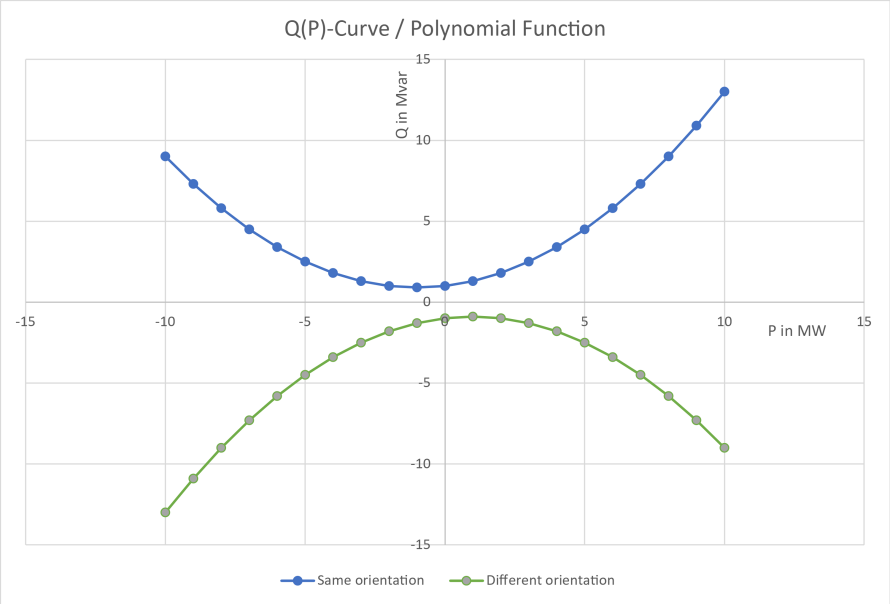


Figure 4.3: Example of a polynomial function with different orientation

5 Application

The Q(P)-Curve object can be selected in different elements when control mode is set to “Q(P)-Characteristic”. The curve is only used for the load flow model (balanced, unbalanced AC load flow).

The following elements are supporting Q(P)-Curve:

- Synchronous machine, 3 phase and single phase model
- Static generator, 3 phase and single phase model
- PV System, 3 phase and single phase model
- Asynchronous machine, configured as DFIG
- Station controller (supported only when values given in Mvar/MW)
- General load, LV load model (3 phase and single phase model)

For further details please see in the technical reference of corresponding model.